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ORIGINAL PAPER

Measuring social value orientation by model-based scoring

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Abstract
This study proposes a method of measuring social value orientation using model-based scoring and a task suitable for such scoring. We evaluated this method by means of parameter recovery simulation (Study 1), and we examined its retest reliability (Study 2) and its predictive validity (Study 3). The results indicate that the proposed method has low bias and sufficient predictive validity. While the improvement in predictive validity of altruism was negligible and comparable to previous scoring methods in terms of confidence intervals, the measurement of equality using the proposed model and task combination produced a moderate correlation that was not observed with other methods. Although SVO is a concept used primarily in psychology, the model assumed in this study is mathematically equivalent to a well-known economics model. We, therefore, suggest that this method may lead to cross-disciplinary research.

Keywords Social value orientation · Social preference · Public goods game · Ultimatum game

1 Introduction

1.1 Cooperation and social value orientation

Decision-makers (DMs) often cooperate without considering their interests even if they do not benefit by cooperating, for example, by helping others, donating blood, and acting in an eco-friendly manner. Social psychologists have long investigated such altruistic behaviors as the study of cooperation. Social psychologists have conceptualized and investigated cooperation through cooperative behaviors when individuals face social dilemmas. Dawes (1980) defined a social dilemma as a situation

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in which individuals in a group have a choice between "cooperating" or "not-cooperating" when it is beneficial for individuals to not cooperate, although it is harmful to the group if everyone chooses not to cooperate. Not cooperating is beneficial for all DMs in social dilemma situations. Therefore, if all DMs want only to benefit their own interest, no one would cooperate. However, some DMs do cooperate in such situations. In addition, individual DMs who face identical situations differ in cooperating or not.

Social value orientation (SVO) can explain the cooperation of DMs in social dilemma situations. Researchers have defined SVO as individual differences in weights attached by DMs when allocating resources to self and others' outcomes (Van Lange 1999; Wyer 1969). SVO theories originated from the interdependence theory of Kelley and Thibaut (1978), who argued that DMs perceive situations of interpersonal interactions subjectively ("effective matrix") rather than objectively ("given matrix"). The literature describes this idea as transforming from a given matrix to an effective matrix. Kelley and Thibaut also noted individual differences in how DMs convert the matrix. Messick and McClintock (1968) adopted this idea to explain cooperation and theorized these individual differences as SVO. DMs who are known as "prosocial" or "altruistic" individuals heavily weigh others' outcomes and tend to cooperate. Numerous studies show correlations between cooperative behaviors and SVO (for reviews, see Au and Kwong 2004; Bogaert et al. 2008; Balliet et al. 2009; Pletzer et al. 2018). Furthermore, it has been suggested that SVO personalities do not make strategic decisions (McClintock et al. 1977) and have high retest reliability (Swap and Rubin 1983; Sheldon 1999).

Because SVO can robustly predict DMs' cooperation, it is worthwhile to predict SVO using other variables (Ackermann et al. 2016). Several experiments have examined how a DM's baseline SVO varies with other characteristics, including in-group or out-group membership, attributes, identity, demographic characteristics, and context. Fleiß et al. (2020), for example, reported that DMs were more altruistic when distributing outcomes between themselves and humanitarian aid NGOs than environmental preservation NGOs. Timilsina et al. (2019a) suggested that there were more prosocial DMs in rural than urban areas, and Jingchao et al. (2021) reported that participants living in plains and hills were more pro-self than those in mountainous areas. Van Doesum et al. (2021) compared SVO in 31 countries and described its variance. Further, Cohen and Hertz (2020) observed that SVO of civil servants changed according to the context (on-duty or off-duty); civil servants were more prosocial when off-duty.

1.2 Methods of measuring SVO

Social psychologists have proposed different methods of measuring individual differences in SVO. The three primary methods include the triple dominance measure (TD), the ring measure (RM), and the SVO slider measure (SSM). The triple dominance measure (TD; Van Lange et al. 1997) is based on Messick and McClintock's (1968) early study of SVO, which categorized DM's motivations into three types: maximizing joint outcome, which was named "prosocial" SVO;

maximizing self-outcome, or “individual” SVO, and maximizing the difference between self and others’ outcomes, or “competitive” SVO. A DM chooses one of these three options for allocating outcomes corresponding to each SVO. The TD consists of nine items. DMs are classified as “prosocial” or “individual” or “competitive”; however, more DMs are categorized as “prosocial” than as "individual" or "competitive," who are often collectively referred to as “pro-self.”

The Ring Measure (RM; Liebrand 1984; Liebrand and McClintock 1988) illustrates the SVO as a vector in a two-dimensional coordinate space. SVO is the vector from the origin to the horizontal (self-outcome) and the vertical axis (others’ outcome). This idea attempts to convert the two indicators weighing the self and others’ outcomes into one index represented by an angle. A DM’s angle is calculated by responses to 24 items. The RM can assess a wider variety of SVOs than the TD. The scale has the ability to categorize participants into the eight categories. However, it is improbable to observe individuals with preferences such as "sadistic" and "masochistic" in empirical settings. Hence, subsequent studies focuses primarily on four categories: "altruistic," "prosocial," "individual," and "competitive".

Murphy and Ackermann (2014) evaluated the TD and the RM according to whether these measures could calculate SVO as a continuous index, whether they could calculate SVO easily using fewer items, and whether the TD and the RM could distinguish inequality avoidance and joint maximization. The literature suggests that SVO should be measured continuously, because the original SVO is a serial concept defined as the weights attached by a DM to self and others’ outcomes. The TD can easily categorize SVO using a small number of items, but it does not calculate SVO continuously. The RM can measure different SVOs, including orientations that weigh negatively on a DM’s and others’ outcomes; however, empirically, we rarely observe a DM having negative preferences. In addition, neither the TD nor the RM can discriminate between inequality avoidance and joint maximization. Eek and Gärling (2006) demonstrated that the two orientations are different, and many DMs prefer inequality avoidance.

Murphy et al. (2011) developed the SSM to address the limitations of the TD and the RM. Each of the six fundamental items of the SSM (Murphy et al. 2011) comprises nine pairs of gains. Each pair consists of one value indicating the amount the decision maker receives ("You receive") and another value indicating the amount an anonymous other receives ("Other receives"). The respondents’ task is to select their preferred pair from the nine available levels. The SVO angle is calculated based on their responses using Eq. 1:

$$\phi_i = \arctan\left(\frac{\overline{x_{oi}}-50}{\overline{x_{si}}-50}\right), \tag{1}$$

where ϕ represents The SVO angle, i represents the respondent, s represents the self, o represents the other, $\overline{x_{si}}$ represents the mean of the selected self-outcome by respondents i, and $\overline{x_{oi}}$ represents the mean of the selected others’ outcomes by respondents i.

Figure 1 illustrates the classification of angles obtained through the SSM for measuring angles. For example, a preference that assigns a weight of 50 to one’s

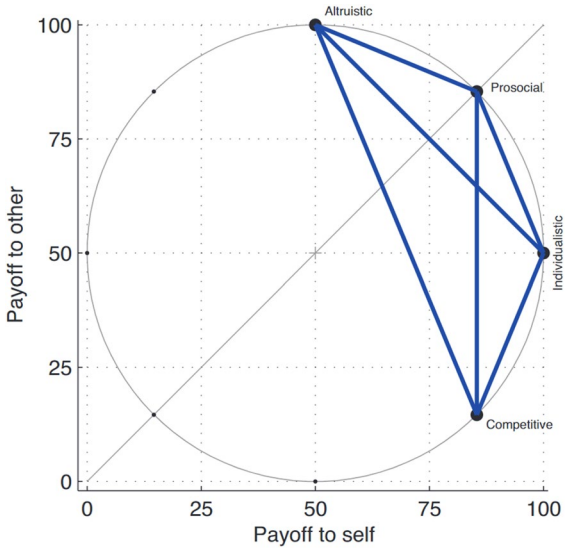

Fig. 1 Conceptual illustration of SVO angles in SSM (Murphy et al. 2011)

own gain and no weight to the opponent's gain (ϕ_i in Eq. 1 equal to 0) is categorized as "individualistic." Likewise, a preference with an ϕ_i of 90 is labeled as "altruistic," while a preference with equal weights of 35 for both gains ($\phi_i = 45$) is termed "prosocial." In Fig. 1, the horizontal axis represents the own gain shown in SSM, while the vertical axis represents the other's gain. The blue line corresponds to each item in SSM.

In addition to the six items used to calculate the SVO angle in the SSM, an additional nine items were prepared by the developers of the measure, resulting in a total of 15 items employed to assess SVO (Murphy et al. 2011; Ackermann and Murphy 2012). As shown in Fig. 1, for SVO to be classified as "prosocial," respondents are required to select pairs that maximize joint benefits while maintaining equal gains. Hence, the six items used to calculate the SVO angle alone do not differentiate whether prosocial respondents are driven by the pursuit of joint maximization or inequality avoidance. The additional nine items are included to discern between these two preferences. Therefore, the combination of the basic six items and the supplementary nine items is deemed sufficient to discriminate between respondents' altruistic tendencies and their concern for equality. This is referred to as the inequality avoidance (IA) index. IA is calculated based on the absolute value of the difference between self and other outcomes (Ackermann and Murphy 2012). The purpose of this index is to determine whether the motivation of "prosocial" decision-makers is driven by the desire to avoid inequality (inequality avoidance) or to maximize joint benefits (joint maximization). IA ranges from 0 to 1, with 0 representing complete inequality avoidance and 1 representing complete joint maximization.

Many recent studies have used the SSM to measure the SVO angle (e.g., Matsumoto et al. 2016; Fiedler et al. 2018; Cohen and Hertz 2020). Mischkowski et al. (2019) measured IA to examine the relationship between equality and HEXACO personality dimensions.

1.3 SVO as a parameter of the social preference model

In addition to the measures described above, SVO is sometimes represented as a utility function, in which a person's utility changes based on the self and others' outcomes (e.g., Wyer 1969). According to Ackermann et al. (2016) and Fleiß et al. (2020), the SSM assumes the following utility function:

$$U_i(x_s, x_o) = x_s + a_i x_o, \tag{2}$$

whereU_i represents DM's own utility, x_s represents the self-outcome, x_o represents the other's outcome, and $a_i \in \mathbb{R}$ represents the weight that DM attaches to others' outcomes. The SVO angle is defined as the weights that DM attaches to the self and others' outcomes. That is, a_i in Eq. 2 corresponds to SVO angle. Further, IA in the SSM represents the tendency of DMs to avoid inequality (equality: $e_i \in \mathbb{R}$). In other words, this model has parameters for the difference between self and others' outcomes. If there is one partner, the utility function is expressed as follows:

$$U_i(x_s, x_o) = x_s + a_i x_o - e_i|x_s - x_o|. \tag{3}$$

We refer to this model as the Ackermann-Fleiß-Murphy Altruism-Equality model (AFM-AE model).

By modifying Eq. 3 to assign a weight of $(1 - a_i)$ to the own gain, it becomes possible to represent a preference that prioritizes the gain of others over one's own gain:

$$U_i(x_s, x_o) = (1 - a_i)x_s + a_i x_o - e_i|x_s - x_o|. \tag{4}$$

The mathematical sociology literature has derived the same utility function as Eq. 4 (Muto 2006), in which parameter a_i is interpreted as individual differences in altruism and parameter e_i is interpreted as individual differences in equality. Muto (2006) demonstrated that this model and the Fehr and Schmidt (1999) model, widely known in economics, are mathematically identical. We use the same notation as Muto (2006) and refer to this model as the Fehr–Schmidt Altruism-Equality model (FS-AE model). In other words, assuming the FS-AE model as the model behind SVO and defining SVO as its parameter allow for the unified interpretation of SVO theory in psychology and social preference theory in economics.

ϕ_i in SSM and the parameters a_i in the AFM-AE model and FS-AE model have different scales. The conversion can be calculated using Eqs. 5 and 6, which correspond to the AFM-AE model, and Eqs. 7 and 8, which correspond to the FS-AE model.

$$a_i = \tan\left(\phi_i \frac{\pi}{180}\right), \tag{5}$$

$$\phi_i = \arctan\left(a_i \frac{180}{\pi}\right), \tag{6}$$

$$a_i = \frac{\tan\left(\phi_i \frac{\pi}{180}\right)}{1+\tan\left(\phi_i \frac{\pi}{180}\right)}, \tag{7}$$

$$\phi_i = \arctan\left(\frac{a_i}{1-a_i}\right)\frac{180}{\pi}. \tag{8}$$

2 Model-based scoring

2.1 Separating models and tasks

The SSM scoring methods may not be ideal for measuring utility function parameters. As shown in Eq. 1, the SVO angle is calculated using the means of selected self-outcomes ($\overline{x_{si}}$) and the means of selected others' outcomes ($\overline{x_{oi}}$). For example, consider a very altruistic DM with an SVO angle of 90°. Using SSM scoring, which averages responses to all items, the measured SVO angle would only be 61.30°. Similarly, for a DM with a very small SVO angle, the SSM measurement would be limited to − 16.26°. Despite the theoretical assumption of a range of − 45° to 90° (Fig. 1), the scoring limitations lead to a smaller range of measurement. This can be problematic, as scores for DMs with preferences outside that range would be unduly centered. Moreover, Bakker and Dijkstra (2021) reported that 99.91% of the respondents who passed the SSM's conscientiousness check (Murphy et al. 2011; Murphy and Ackermann 2014) were classified as either "prosocial" or "individual." In summary, distribution of SVO angles tends to cluster around the center.

We propose a new method of calculating SVO by estimating the parameters of the utility function: model-based scoring. In this method, the parameters of the AFM-AE model and the FS-AE model are estimated from DM's choices. The estimated value is considered to be each DM's SVO. This method has two advantages: it allows us to calculate the SVO without averaging and to estimate equality independently of altruism.

We can apply model-based scoring to data from a variety of tasks. Preparing a probability model adjusted to the task's characteristics (i.e., the number of choices) allows us to estimate the parameters. For example, a DM chooses one of nine pairs of choices in the SSM for his or her and others' outcomes. The following probability model fits this data. We assume that respondent i's response to the slider task for item t, $\text{choice}_{it} \in \{1, \ldots, 9\}$ follows a categorical distribution, such that

$$\text{Choice}_{\text{it}} \sim \textit{Categorical}\left(\theta_{it}\right), \tag{9}$$

and $\theta_{it} \in (0, 1)$ is a nine-dimensional vector representing the probability of choosing each of the nine choices in item t for respondent i. We assume that $\theta_{it} \in (0, 1)$ is calculated by the softmax function

$$\theta_{itk} = \frac{\exp(U_{itk})^{\lambda_i}}{\sum_{k=1}^{9}\exp(U_{itk})^{\lambda_i}}, \tag{10}$$

where λ_i is the rationality parameter. When $\lambda_i = 0$, each option is chosen at random. On the other hand, if $\lambda_i = \infty$, the option with high utility is always preferred if there is even a small difference in utility. If we substitute self-outcomes $x_{s,t}$ and other's outcomes $x_{o,t}$ presented in each item into Eq. 3 or Eq. 4, we obtain the utility U_{it} for the choice $k \in \{1,\ldots,9\}$:

$$U_{itk} = x_{s,tk} + a_i x_{o,tk} - e_i \left| x_{s,tk} - x_{o,tk} \right|. \tag{11}$$

$$U_{itk} = \left(1 - a_i\right) x_{s,tk} + a_i x_{o,tk} - e_i \left| x_{s,tk} - x_{o,tk} \right|. \tag{12}$$

The a_i estimate refers to the respondent's altruism, and e_i refers to the respondent's equality. $x_{s,tk}$ is the kth self-gain presented during trial t, while $x_{o,tk}$ signifies the k th counterpart gain in trial t. Since the same items are presented to all respondents i, both x_s and x_o remains unchanged with respect to i.

2.2 A new task to measure SVO

Because the 15 items of the SSM may be insufficient to estimate the parameters, we prepared a new task better suited to parameter estimation. This task required two easy choices, although the respondents had to respond to more items. In constructing the tasks, we referred to psychophysical measurement methods and methods of judging equality equivalence points, i.e., points that DM perceives to be equivalent (Kerschbamer 2015). In psychophysical measurement methods, respondents are asked to report a quantity they feel is subjectively equivalent to some physical quantity. A typical example of this task is the delayed value discount task (Rachlin et al. 1991). We named our task the Equal Equivalent Measure (EEM), because we asked the respondents to evaluate equality equivalence.

The respondents chose between two options for distributing rewards between themselves and their partners in this task. The option shown on the left-hand options present an equal distribution, and the option on the right presents an unequal distribution—the amount of money on the left decreases from 900 to 0 yen in 10 steps. Because people generally tend to dislike unequal distributions, respondents were initially expected to choose the option on the left and then, at some point, choose the one on the right. The pattern of the amount of money on the right-hand options change after reducing the amount of the option on the left to 0 yen, such that there were four patterns for the right-hand option: "900 yen for you, 500 yen for A," "700 yen for you, 500 yen for A," "300 yen for you, 500 yen for A, "and "100 yen for you, 500 yen for A." A was an anonymous respondent who was a stranger without any expected future relationship. Because it was impossible to determine whether a respondent who chose the left-hand option for all the choices was a complete egalitarian or answered insincerely using only these items, we provided 12 items

for which the respondents had to choose between unequal distributions in which the sum of the amounts was equal. As a result, this task consisted of 52 items.

We estimated the parameters from the respondent's choices to the items using the probability model described below. The choice of a respondent i to each item t follows a Bernoulli distribution, because the data are binary:

$$Choice_{\mathrm{it}} \sim Bernoulli\left(\theta_{it}\right). \tag{13}$$

The probability $P(Choice_{it} = 1)$ of choosing the option on the right is denoted as θ_{it}, where θ_{it} is represented by the sigmoid function

$$\theta_{it} = \frac{\exp\left(U_{it}{}^{\mathrm{right}}\right)^{\lambda_i}}{\exp\left(U_{it}{}^{\mathrm{right}}\right)^{\lambda_i} + \exp\left(U_{it}{}^{\mathrm{left}}\right)^{\lambda_i}} = \frac{1}{1+\exp\left(-\lambda_i\left(U_{it}{}^{\mathrm{right}} - U_{it}{}^{\mathrm{left}}\right)\right)}\# \tag{14}$$

that expresses the probability of one of the two options being chosen (Nilsson et al. 2011). Substituting self-outcomes $x_{s,t}$ and other's outcomes $x_{o,t}$ presented for each item into Eq. 5, we obtain

$$U_{it} = \left(1 - a_i\right)x_{s,t} + a_i x_{o,t} - e_i\left|x_{s,t} - x_{o,t}\right|.\# \tag{15}$$

2.3 Hierarchical Bayesian estimation

We used the hierarchical Bayesian method to estimate each respondent's parameters in model-based scoring. One method of estimating each respondent's parameters is to estimate independent individual parameters; however, this method requires estimating many parameters, resulting in poor accuracy. The hierarchical Bayesian model method assumes that an individual's parameters are generated from a group-level distribution. As a result, the group information can be used to estimate an individual's parameters. These parameters can be estimated for each DM, thus improving the estimation accuracy (Nilsson et al. 2011). Therefore, using a hierarchical Bayes model efficiently estimates altruism and equality in the AFM-AE model and the FS-AE model. The hierarchical model for this estimation is as follows:

$$\begin{gathered}
\begin{pmatrix} a_i \\ e_i \end{pmatrix} \sim \mathrm{MultiNornal}(\boldsymbol{\mu}, \Sigma), \\
\sigma_{ae} = \sigma_a \sigma_e \rho_{ae}, \\
\begin{aligned} \mu_a \mu_e &\sim Normal(0, 10), \\ \sigma_a \sigma_e &\sim Exponential(1.0), \\ \rho_{ae} &\sim LkjCorr(1), \# \end{aligned} \\
\lambda_i \sim \mathrm{LogNormal}\left(\mu_\lambda, \sigma_\lambda\right),
\end{gathered} \tag{16}$$

$$\mu_\lambda \sim \text{Normal}(0, 10),$$

$$\sigma_\lambda \sim \text{Exponential}\left(\frac{1.0}{2.5}\right).$$

Here, μ represents the mean, σ stands for the standard deviation, σ_{ae} denotes the covariance matrix of a and e, ρ_{ae} is the correlation coefficient of a and e, and σ_{ae} corresponds to the component of Σ. Given that Stan suggests employing the LkjCorr prior distribution for correlation coefficients, this study adopted this recommendation (Stan Development Team 2022). LkjCorr(1) in Eq. (16) is equivalent to a uniform distribution within the range of -1 to 1.

3 Study 1: parameter recovery simulation

Study 1 conducted parameter recovery simulations to evaluate the performance of model-based scoring, in order to determine whether the correlation between true values generated by the FS-AE model simulation was the true model and to estimate values scored by the SVO angle and model-based scoring. We also examined whether the SSM's scoring method resulted in scores biased towards the center by generating SSM data and checking the correlation between the SVO angles and the true values. Further, we tested the possibility of calculating altruism a_i and equality e_i from the SSM data by model-based scoring using hierarchical Bayesian estimation without SSM scoring. If this was successful, we then expected to conduct a new analysis by calculating altruism a_i and equality e_i from the existing SSM data.

3.1 Methods

The parameter recovery simulation[1] method consisted of the following steps: (1) generating random numbers (true values) for a_i and e_i using the AFM-AE model and FS-AE model as the true model; (2) generating the SSM and the EEM data by substituting the generated true values into the probability model; (3) calculating indicators from those data (estimated values); and (4) checking the correlation between estimated values and true values. The simulation consisted of 100 simulations trials with a sample size of 200 for each trial. The altruism and equality parameters were generated by random numbers from a normal distribution using R's rnorm function, with a mean of 0.5 and a standard deviation of 0.5. We set the rationality parameter λ_i as 1. We used variational Bayesian inference by Stan (Carpenter et al. 2017) for this estimation. Variational Bayesian inference is a fast estimation method with the constraint that the posterior distributions of the parameters are independent.

[1] All simulations and analyses in this article were conducted using R 3.6.3 (R Core Team 2019).

Table 1 Correlation coefficient between true and estimated values

True model	Task	Scoring	Angle	Altruism	Equality
AFM-AE Model	SSM	Original	0.67 (7.59)*	–	–
FS-AE Model	SSM	Original	0.78 (0.35)*	–	–
AFM-AE Model	SSM	Model-based	–	0.87 (0.24)	0.94 (0.17)
FS-AE Model	SSM	Model-based	–	0.89 (0.22)	0.89 (0.26)
AFM-AE Model	EEM	Model-based	–	0.90 (0.21)	0.69 (0.35)
FS-AE Model	EEM	Model-based	–	0.96 (0.13)	0.93 (0.18)

Values in parentheses represent root mean squared error (RMSE)

*To standardize the scale with other indicators, transformations were applied using Eq. 5 or 7

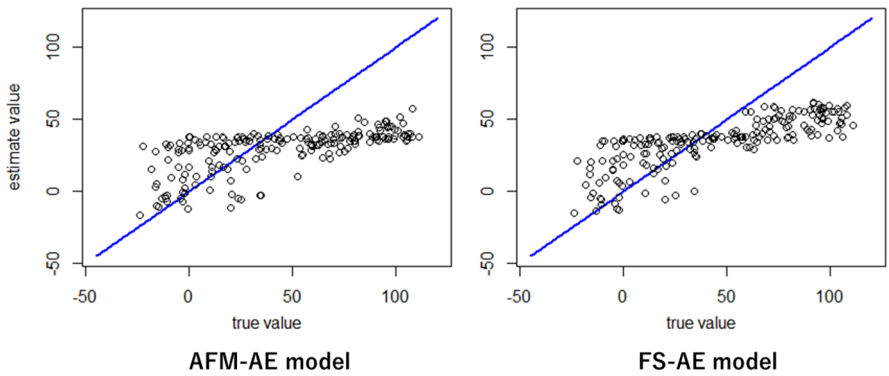

Fig. 2 Scatter plots of true and estimated values from SSM's original scoring

3.2 Results

We calculated the correlation between estimated and true values. The scoring and correlation coefficient calculations were conducted separately for each simulation. The mean of the results of 200 simulations is reported in Table 1.

Figure 2 displays a scatter plot of the estimated and true SVO angle values calculated using the original scoring of the SSM.

Figure 3 presents scatter plots of altruism and equality that illustrate the relationship between true values and estimated values obtained through model-based scoring, using the same SSM data as in Fig. 2.

3.3 Discussion

The original SVO scoring exhibits significant score bias, whether the assumed model is the true model of the AFM-AE model or the FS-AE model, an extension introduced in this study. As an illustration, Fig. 2 shows that the scores cluster around 45° (reflecting prosocial tendencies) even for decision makers whose true

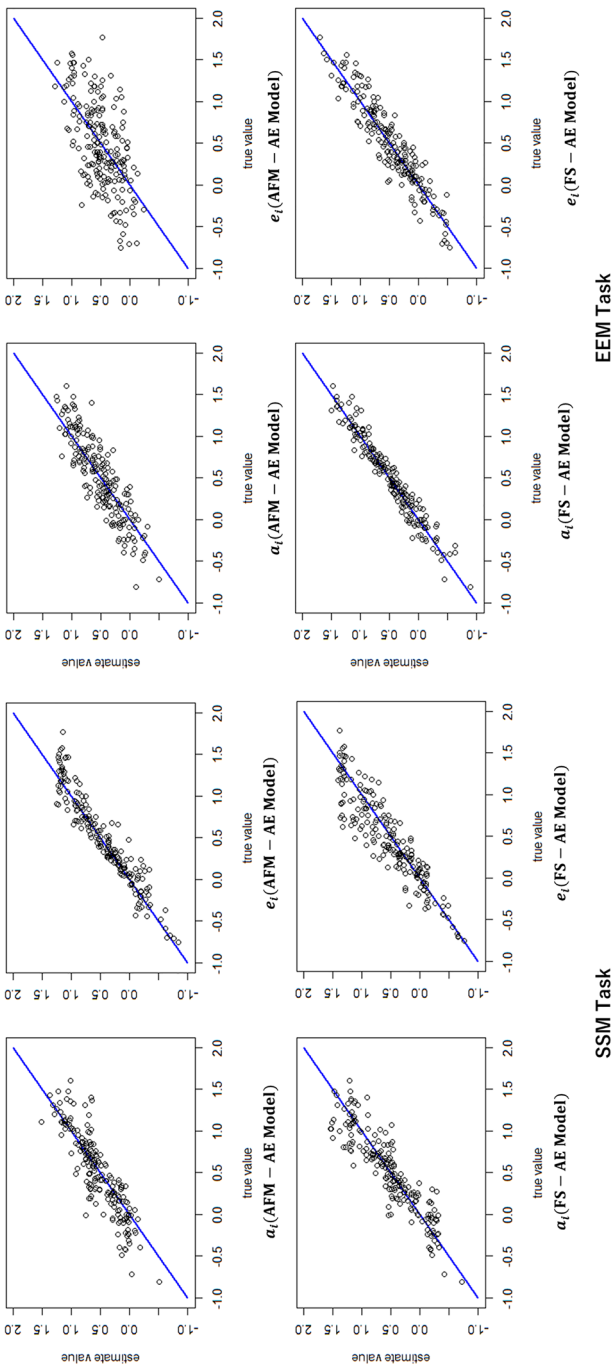

Fig. 3 Scatter plots of true and estimated values from model-based scoring

values are at 0° (indicating complete individualism) and 90° (representing altruism). This observation implies that the original scoring method, involving averaging of responses to all items, may have introduced scoring bias.

Model-based scoring demonstrated a reduction in bias. For the identical SSM data that displayed notable bias in the original scoring, bias was markedly reduced with model-based scoring, whether the true model was the AFM-AE model or the FS-AE model (Fig. 3). This outcome implies the potential for bias correction through model-based scoring, even with previously obtained SSM data.

For model-based scoring within the context of the SSM task data, both the altruism parameter ($r = 0.87$) and the equality parameter ($r = 0.94$) exhibit strong correlations with the true value when the AFM-AE model, the same model assumed in the original study, is considered as true (as presented in Table 1). On the other hand, using the FS-AE model as the true model yields a slightly higher correlation for the altruism parameter and the true value ($r = 0.89$), but a lower correlation for the equality parameter ($r = 0.89$), compared to the scenario with the AFM-AE model as the true model. This suggests that the design of the SSM task may have been more aligned with the initially assumed AFM-AE model.

For the EEM task data, both the altruism parameter ($r = 0.90$ vs. $r = 0.96$) and the equality parameter ($r = 0.69$ vs. $r = 0.93$) exhibit higher correlations with true values when the true model is the FS-AE model than when the AFM-AE model is considered the true model. The differences were particularly pronounced for the equality parameter. The combination of the FS-AE model and the EEM task displayed lower bias than all other combinations. These findings suggest that different models may be better suited for different tasks, and the FS-AE model combination demonstrates the least bias among the combinations examined in this study.

4 Study 2: comparing models and retest reliability with real data

This section presents the outcomes of a model comparison between the AFM-AE model and its extension, the FS-AE model, to determine their suitability for empirical data. Note that this comparison of empirical fit was not preplanned but rather conducted post hoc.

The primary goal of this study was to investigate retest reliability. SVO is a stable individual trait that is used to predict cooperation. To satisfy this property, SVO scores must be consistent over time. Several studies have demonstrated the high retest reliability of SVO angles. Murphy et al. (2011) measured SVO angles one week apart and reported a correlation of $r = 0.92$ between angles. Bakker and Dijkstra (2021) reported a correlation of $r = 0.60$ in a three-month apart design. De Matos Fernandes et al. (2022) measured the SVO angle over six months using monthly measurements and reported an intra-class correlation of $\rho = 0.78$. A correlation between scores as high as that for the SVO angle would indicate that the parameters have sufficient test–retest reliability. However, to our knowledge, no studies have examined the test–retest reliability of IA. To fill this gap, we conducted two surveys to examine the retest reliability of parameter estimated model-based scoring, in which we measured IA twice with an interval between the measurements.

4.1 Method

4.1.1 Participants

We distributed a URL linked to questionnaires created in Qualtrics, an online survey platform, to students in a social psychology lecture course. The identical survey was conducted twice at an approximate interval of two months. There were 474 respondents to the first survey and 432 respondents to the second survey, and 386 participants who completed both surveys and responded to the Directed Questions Scale (DQS; Maniaci and Rogge 2014) exactly as instructed were considered for the analysis (133 men, mean age 19.60 years, SD = 1.33).

4.1.2 Survey items

The questionnaires consisted of the Psychological Entitlement Scale (Campbell et al. 2004) translated into Japanese (Shimotsukasa and Oshio 2016), the EEM, the TD (Van Lange et al. 1997), and the SSM (Murphy et al. 2011). We also inquired about the respondents' age and gender. We included the DQS with the Japanese version of the Psychological Privilege Scale items for Satisficer detection (Miura and Kobayashi 2015). The respondents were instructed to choose "not very applicable" in the DQS. The participants who did not respond as instructed were considered to have failed to read the question carefully and were excluded from the analysis.

4.2 Results

4.2.1 Model comparison

To determine whether the AFE-AE model or the FS-AE model provides a more accurate empirical explanation, we compared the models using log-likelihood. The results indicated that the FS-AE model offered a better fit to the data, both for the SSM task (− 4853 vs. − 4112) and the EEM task (− 7926 vs. − 6620). Consequently, in the subsequent sections discussing model-based scoring, we present parameter estimates obtained from fitting the FS-AE model.

4.2.2 SVO angle

The SVO angle was calculated from the SSM data according to Murphy et al. (2011). We also calculated IA of individuals classified as prosocial. The mean SVO angle in the first survey was 24.14° (SD = 16.66). Therefore, we classified 256 of the 386 individuals as prosocial. The correlation between SVO angle and altruism a_i was $r = 0.59(95\%\text{CI}[0.52, 0.65])$; the correlation between IA and equality e_i was $r = -0.24(95\%\text{CI}[-0.35, 0.12])$; and the correlation between SVO angle and IA was $r = -0.28(95\%\text{CI}[-0.39, -0.16])$.

The mean SVO angle in the second survey was 26.81° (SD = 15.00). The number of individuals classified as prosocial was 255 of 386. The correlation between the SVO angle and altruism a_i was $r = 0.59$(95%CI[0.52, 0.65]); the correlation between IA and equality e_i was $r = -0.24$(95%CI[−0.35, −0.12]); the correlation between SVO angle and IA was $r = -0.28$(95%CI[−0.39, −0.16]); and the correlation between SVO angles in the first and second surveys was $r = 0.67$(95%CI[0.62, 0.73]).The correlation between IAs in the first and the second survey was $r = 0.39$(95%CI[0.29, 0.50]).

4.2.3 Parameters estimated by model-based scoring

The estimation of parameters was conducted using Stan (Carpenter et al. 2017). The code for estimation was identical to that used in Study 1.

4.2.3.1 SSM The mean altruism parameter for the SSM data among individuals in the first survey was − 0.38 (SD = 0.82), and the mean equality parameter for the SSM data was 0.80 (SD = 0.12). The correlation between altruism and equality was $r = 0.61$(95%CI[0.55, 0.67]). The mean altruism parameter among individuals in the second survey was − 0.25 (SD = 0.44), and the mean equality parameter value was − 0.15 (SD = 0.44). The correlation between altruism and equality was $r = 0.61$(95%CI[0.54, 0.66]). The correlation of the altruism parameter between the first and second surveys was $r = 0.62$(95%CI[0.57, 0.69]) The correlation of the equality between the first and second surveys was $r = 0.53$(95%CI[0.48, 0.62]). The Japanese version of the psychological sense of privilege was negatively correlated with altruism, with coefficients of $r = -0.24$(95%CI[−0.33, −0.15]) for the first survey and $r = -0.29$(95%CI[−0.39, −0.21]) for the second survey; this was also negatively correlated with equality, with coefficients of $r = -0.22$(95%CI[−0.32, −0.13]) for the first survey and $r = -0.23$(95%CI[−0.32, −0.13]) for the second survey.

4.2.3.2 EEM The mean altruism parameter for the EEM data among individuals in the first survey was 0.33 (SD = 0.38), and the mean equality parameter for the EEM data was 0.80 (SD = 0.84). The correlation between altruism and equality was $r = 0.21$(95%CI[0.12, 0.31]). The mean of altruism a_i among individuals in the second survey was 0.36 (SD = 0.30) and the mean equality was 0.60 (SD = 0.66). The correlation between altruism and equality was $r = 0.18$(95%CI[0.09, 28]). The correlation of the altruism parameter between the first and second surveys was $r = 0.65$(95%CI[0.59, 0.70]), whereas the correlation of equality between the first and second surveys was $r = 0.53$(95%CI[0.45, 0.60]). The Japanese version of the privilege scale showed a negative correlation with altruism, with coefficients of $r = -0.22$(95%CI[−0.31, −0.12]) in the first survey and $r = -0.28$(95%CI[−0.37, −0.18]) in the second survey; for equality, there was a weak or slight negative correlation, with coefficients of $r = -0.09$(95%CI[−0.19, 0.01]) in the first survey and $r = -0.12$(95%CI[−0.23, −0.03]) in the second survey.

4.2.3.3 Correlation between EEM and SSM The correlation between altruism estimated from SSM and altruism estimated from EEM was $r = 0.62(95\%\text{CI}[0.55, 0.67])$ for the first survey and $r = 0.70(95\%CI[0.66, 0.76])$ for the second survey. For equality, the correlation was $r = 0.17(95\%\text{CI}[0.08, 0.27])$ for the first survey and $r = 0.21(95\%\text{CI}[0.11, 0.30])$ for the second survey.

4.3 Discussion

We collected data from two surveys conducted over 2 months. Model comparisons based on log-likelihood between the AFM-AE model and its extension, the FS-AE model, demonstrated that the FS-AE model provided a better fit to the data, whether derived from the SSM or the EEM. The distinction between these models lies in whether altruism is primarily a weight attributed to the other's gain as expressed in Eq. 2 or a parameter reflecting the equilibrium between self and other gains in Eq. 3. These results suggest that characterizing altruism through the equilibrium of self-other gains is more empirically aligned with the data as a model of human decision-making in reward distribution.

Our main purpose was to investigate retest reliability by examining the correlations between the first and second surveys. The results indicated that the SVO angle was slightly higher than the EEM for the altruism index (the estimate of altruism parameter in the EEM and the SVO angle in the SSM). However, the two measures had similar retest reliability. On the other hand, neither the equality measure estimating altruism in the EEM nor IA in the SSM was high.

The lower correlation between the retest reliability of the SVO angle in these two surveys, compared to those in Murphy et al. (2011) and Bakker and Dijkstra (2021), may be explained by the longer duration between surveys in the present study. Nevertheless, the present results indicate that altruism scored by model-based scoring has retest reliability comparable to the SVO angle.

The retest reliability of equality indicators in the present study was not high for either measure. Specific studies have suggested that individuals' SVO angle changes over time and across situations (Anderl et al. 2015; Cohen and Hertz 2020). However, there is currently no literature on changes in IA. Moreover, the results of this study did not clarify whether there were problems with the measurement method or whether equality is more changeable than altruism. Therefore, further investigations of this issue are warranted.

5 Predictive validities

5.1 Study 3-1: public goods game

We examined the predictive validity of the altruism parameter using the public goods game, also known as a social dilemma game, in which an individual benefits from not cooperating regardless of the other party's behavior. It has been

Table 2 Rank correlation coefficient between cooperation in the PGG and each index

	SVO angle	Model-based scoring	
		SSM	EEM
First period	0.21 [0.00 0.40]	0.24 [0.03 0.43]	0.26 [0.05 0.45]
Last period	0.24 [0.03 0.43]	0.28 [0.07 0.46]	0.33 [0.13 0.50]

shown that individuals with more altruism will cooperate more in games with this structure (e.g., Bogaert et al. 2008). Thus, we expected that DMs with larger SVO angles and altruism parameters estimated by model-based scoring would tend to make larger contributions.

5.1.1 Methods

5.1.1.1 Participants We recruited participants in the same social psychology course as in Study 2, but two months later. A total of 88 students participated (26 men, mean age 19.80 years, SD = 1.53).

5.1.1.2 Procedure This experiment was conducted via Zoom, an online video conferencing tool. The names of the participants were randomized by the experimenter such that the participants were anonymous. We briefed the participants on the experimental procedure and the rules of the public goods game (PGG). They then responded to a test to confirm their understanding of the rules and played 20 repetitions of the PGG. We did not inform the participants about the number of repetitions. At the end of the experiment, the participants received a reward in exchange for the points they earned in the PGG, with each point calculated as 1 yen. We sent the rewards to the participants by email in the form of Amazon gift cards.

The standard procedure of the public goods game was conducted, in which the participants were given 20 points by the experimenter at the beginning of a period. We asked the participants to decide how many of these 20 points they would contribute to the group. The sum of the points contributed by all participants was multiplied by 1.6 and divided by 4. Each participant was given feedback on the number of points they earned during the period.

5.1.2 Results

The mean contribution by participants across the 20 periods was 7.21 (SD = 4.44). The mean of the first contribution was 9.57 (SD = 4.89), and that at period 20 was 5.97 (SD = 5.52). Table 2 shows the correlation between each indicator and contribution. The numbers in square brackets represent the 95% confidence intervals.

5.1.3 Discussion

We examined whether the altruism parameter estimated by model-based scoring predicted cooperation in a PGG with incentives. The results indicated that the altruism parameter predicted the participants' contribution. In previous SVO studies (Murphy et al. 2011), a correlation of approximately $r = -0.24$ was considered to have sufficient predictive validity for cooperation in prisoner's dilemma games. Therefore, we concluded that the altruism parameter estimated using this method has predictive validity.

5.2 Study 3-2: ultimatum game

We used the ultimatum game (UG) to examine the predictive validity of the equality parameter. In this game, it is rational for senders to accept an unequal proposal if they consider only their outcomes. However, the sender may reject an unequal proposal due to his or her motivation against inequality. In this study, we used a strategy method in which the receiver could decide among all possible proposals that the sender chose to send. People with high equality were expected to have a higher probability of rejecting proposals other than equal distributions, resulting in a lower overall acceptance rate. Therefore, we expected that people with higher equality would have a lower acceptance rate.

5.2.1 Methods

5.2.1.1 Participants Participants were recruited in a social psychology class. We sent a URL linked to a questionnaire created in Qualtrics to the students who agreed to participate. The participants ($n = 108$, 25 men, mean age 19.37 years, SD = 1.12) who correctly answered fewer than 4 of 8 quiz questions on the rules of the ultimatum game were excluded from the analysis. The data from a total of 107 participants were analyzed.

5.2.1.2 Procedure The participants were asked to complete an online survey during a 3-day response period. We randomly matched the responses with each other at the end of the response period and calculated the rewards. The money earned in the survey was given directly to the participants as rewards sent via email in the form of Amazon gift cards.

The questionnaires included the EEM, the SSM, and the ultimatum game. We explained the rules of the ultimatum game to the participants, followed by a quiz to confirm their understanding of the rules. They then saw a webpage for making decisions as a sender and a webpage for making decisions as a receiver. After deciding as a sender, the participants had to choose whether to accept or reject all the proposals that the sender chose. Thirteen distribution levels were provided in increments of 50 yen, ranging from 600 yen to the sender and 0 yen to the receiver, and 0 yen to the sender and 600 yen to the receiver. The questionnaire repeatedly emphasized that the money obtained in the ultimatum game would be received as the actual reward.

5.2.2 Results

We conducted GLMM by assuming a binomial distribution. The objective variable was how many of the 12 types of distributions other than the equal distribution were accepted. Explanatory variables were standardized. The intercept was the variable effect. The IA estimate was $\beta = -0.04[-0.34, 0.25]$. The equality parameter estimated from the SSM data with model-based scoring was $\beta = 0.05[-0.16, 0.27]$, and that estimated from the EEM data was $\beta = -0.28[-0.48, -0.08]$.

5.2.3 Discussion

Study 3-2 examined whether equality scored by the SSM and the EEM predicted the acceptance rates in an incentivized ultimatum game. Participants with higher equality parameters measured by the EEM were more likely to reject proposals other than the equality proposal. This result indicates that the equality parameter as measured by the EEM does not measure a marked deviation from individual differences in the tendency to avoid inequality. On the other hand, IA and equality parameters measured from the SSM data did not predict rejection by DMs. In the EEM, DMs always compared equal distributions versus unequal distributions, whereas in the SSM, DMs did not always compare them. In short, choosing an equal distribution in the EEM task meant rejecting unequal alternatives. This difference may explain the differences between the EEM and the SSM in predicting rejections.

6 General discussion

This study proposed a new method of estimating the model parameters assumed in SVO, which we named model-based scoring. We conducted SVO slider measure and equal equivalent measure methods in three studies. Study 1 revealed that the method used to calculate the SVO angles, when compared to the true value, created a bias that could be reduced by model-based scoring. Study 2 indicated that the SVO model (FS-AE model) extended in this study aligned well with empirical data, and altruism parameters estimated by model-based scoring had better retest reliability than SVO angles. On the other hand, none of the indicators related to inequality avoidance had high retest reliability. In Study 3, the altruism parameter estimated by the SSM and the EEM significantly predicted cooperation in PGGs with incentives. In addition, equality estimated from the EEM predicted acceptance rates in ultimatum games, whereas equality estimated from the SSM did not predict these rates.

6.1 Advantages of model-based scoring in measuring SVO

The most significant contribution of this study is the introduction of model-based scoring to measure SVO. In other SVO measurements, such as SSM, a model of social preference is assumed, and SVO is defined as its parameter (Ackermann et al.

2016; Fleiß et al. 2020). However, the conventional scoring method does not directly measure that parameter. Specifically, scoring by adding and averaging respondents' responses to all items is problematic because it tends to skew the scores of respondents whose responses are inconsistent towards the middle (Bakker and Dijkstra 2021). Moreover, it does not measure the theoretically assumed range of − 45°–90° but instead measures from − 16.26° to 61.26°, causing a limitation in accurately capturing SVO as theoretically intended. Study 1 demonstrated that conventional scoring still exhibits bias, clustering scores near the center. In contrast, model-based scoring shows reduced bias, even when applied to the same SSM data. One reason for the smaller bias in model-based scoring is that it permits measurement of a wider range of SVOs, because there is no restriction to average responses to all items. Additionally, while SSM has six items to measure SVO and nine items to discriminate whether participants classified as prosocial are egalitarians or joint interest maximizers, model-based scoring enables the use of all 15 items for scoring, thereby improving reliability.

With model-based scoring, it is possible to assume various social preference models and compare their empirical fits. In this study, we considered the FS-AE model, which is an adjusted version of the model assumed in the SSM. Adopting the FS-AE model for measuring SVO offers both theoretical and empirical advantages. Theoretically, the FS-AE model has been demonstrated to be mathematically equivalent to Fehr and Schmidt's (1999) inequality avoidance social preference model, widely used in economics (Muto 2006). By defining SVO as individual differences in its parameters, the SVO theory in psychology and the social preference theory in economics can be unified, facilitating connections between these fields. Empirically, as shown in Study 2, the FS-AE model provides a better fit to the data compared to the model assumed in the SSM, whether the data are from the SSM or the EEM. This suggests that the FS-AE model more accurately represents respondents' decision-making in the money distribution task.

Furthermore, model-based scoring allows for the separation of the model from the task, in order to determine which tasks align best with the model. In this study, EEM was introduced as a new task. EEM uses a paired comparison method; a similar task that has been used in many research fields (e.g., Beck and Triplett 2009) is the delay discounting task (Rachlin et al. 1991). In this task, DMs choose between getting less money now or more money later. The choice made in this task indicates the equivalence point where the two options have an equal value for the DM. Similarly, the EEM task asks DMs to provide their equivalence points for equal and unequal distribution values to estimate the unbiased equality parameter. The EEM requires many items to ensure its reliability; however, the EEM does not necessarily require 52 items, as in this study, because the number of items can be decreased to reduce the questionnaires' length or the respondents' burden. The EEM requires a set of items with a larger distribution than the partner's items (e.g., self: 900, partner: 500) and a set of items with a smaller distribution than the partner's items (e.g., self: 300, partner: 500). Nevertheless, the researcher can increase or decrease the interval between the number of options according to the desired reliability. Overall, the present study demonstrated that EEM is an effective method for measuring the parameters of the FS-AE model. The combination of the FS-AE model and EEM exhibited the least bias (Study 1). Moreover, the

examination of predictive validity in Study 3 revealed that the altruism parameter had the highest value in this combination. Only equality as measured by this combination significantly predicted acceptance/rejection in the ultimatum game.

By combining the FS-AE model and EEM, we were able to score equality parameters accurately, overcoming the limitation of previous research that the equality orientation of all respondents could not be measured. The original scoring of the SSM measured respondents' propensity to avoid inequality as an indicator called IA. IA is indexed in the SSM to identify whether DMs grouped as prosocial are motivated to avoid inequality or maximize joint outcomes. Therefore, IA can be calculated only for DMs whose SVO angle is classified as prosocial. An additional nine items are required to measure IA. Many studies, however, either did not measure IA using the additional nine items, or they used only six main items (Anderl et al. 2015; Thielmann and Böhm 2016; Mischkowski and Glöckner 2016; Bieleke et al. 2017; Fiedler et al. 2018; Haesevoets et al. 2020). In addition, Van den Bergh et al. (2006) reported that equality orientation is not unique to prosocial individuals but also common to pro-self individuals. Further, Mischkowski et al. (2019) indicated that IA could calculate the equality orientation only of people classified as prosocial, which was a limitation of their study. These results suggest that it would be worthwhile to measure the equality orientation of DMs who are not classified as prosocial.

The experimental significance of measuring altruism and equality through the EEM task + FS-AE model resides in the distinct identification and quantification of altruism (favoring the other party in gain trade-offs) and equality (disfavoring disparities in gains, regardless of direction). Notably, both parameters can be extracted from the same behavioral data. This presents an intriguing possibility: unraveling the relative influence of altruism and equality in previous findings that associate SVO with specific behaviors or modulate the impact of one variable on the magnitude of another.

Model-based scoring can also offer a solution for dealing with respondents who show inconsistency in their responses. Equations 10 and 14 are often used in reinforcement learning (softmax action selection) as models of DMs' choice behavior mechanisms. In reinforcement learning, λ_i in the formulas is interpreted as an exploration parameter (Katahira 2015; Erev and Barron 2005). The larger the λ_i, the more DMs select options according to their values; when λ_i is 0, DMs make entirely random selections. In this study, the λ_i parameter was considered the rationality parameter, and DMs with low rationality parameters responded inconsistently. Therefore, the method proposed in this study allows us to estimate altruism and equality by the reduction of noise in the consistency of responses. The λ_i has the characteristic of decreasing when DM's choices become increasingly inconsistent. Therefore, we can also exclude DMs with significantly small λ_i from analysis. Further investigations are needed to develop standards for judging if DMs make arbitrary decisions.

6.2 SVO: continuity and typology

As pointed out by Murphy and Ackermann (2014), if SVO is defined as individual differences in weighting of self- and other-gains, it should inherently be continuous. Empirical evidence also supports its continuous, rather than categorical, nature (Fiedler

et al. 2013). Yet SVO is frequently utilized to categorize participant types. For example, several research practices classify participants into prosocial and egocentric groups, using SVO as a regulating variable (e.g., Kornhauser and Tontrup 2020; Kuss et al. 2015; Yamagishi et al. 2017). Our proposed method does not exclude the use of SVO for typing, as Murphy et al. (2011) provide criteria for typing respondents based on the angle measured by SSM, although this has been deprecated. Similarly, it may be possible to use the parameters measured by model-based scoring for typing, as has been done in the past, if suitable criteria are established in the future. One possible criterion could involve converting the altruism parameter values to SVO angles on a scale using Eq. 6 or 8 and applying the same criteria used by Murphy et al. (2011) for classification.

6.3 Significance of precise measurement

This study contends that measuring SVO without bias is essential. The development of the SSM (Murphy et al. 2011) has enabled the measurement of SVO as a continuous variable. Consequently, an increasing number of studies now treat SVO as a dependent variable and explore its variance (Ackermann et al. 2016; Cohen and Hertz 2020; Van Doesum et al. 2021; Fleiß et al. 2020; Timilsina et al. 2019b; Jingchao et al. 2021). In such research, measuring SVO with less bias is of great importance.

Although the minor bias observed in the original model-based scoring of Study 1 demonstrated enhanced predictive capacity for altruism in Study 3-1, the comparison of confidence intervals revealed no statistically significant distinctions in the context of altruism. In terms of cooperation in public goods games, the predictive capability of model-based scoring showed a slight elevation of approximately 0.03. We think that this minor uptick does not yield a fundamentally novel insight into human psychology, given the myriad factors influencing decision-making in public goods games, coupled with the minimal magnitude of the increase. However, it is important to recognize that this seemingly modest increase in predictive power could have notable implications when applied to real-world psychology data. Specifically, a variance of 0.03 could translate into significant adjustments in required sample sizes during effect size-based sample size calculations. From this standpoint, the potential to achieve more precise measurements could potentially lead to substantial reductions in necessary sample sizes, a consideration we deem far from insignificant.

Conversely, within the realm of equality (Study 3-2), we observed moderate correlations aligned with the theoretically anticipated direction only when utilizing the FS-AE model + the EEM task combination introduced in this study. In contrast, other methods exhibited minimal correlations. These outcomes highlight the heightened efficacy of the method proposed in this study, particularly in terms of equality.

6.4 Substantial meaning of altruism and equality

Lastly, we delve into the substantial meaning of altruism and equality. In conjunction with these parameters and other psychological variables, Study 2 revealed a noteworthy negative correlation between scores on the self-reported psychological

sense of privilege scale and altruism as quantified by the FS-AE model + the EEM task Conversely, regarding equality, the current findings do not allow us to definitively ascertain the presence of either a negative correlation or no correlation. The observation that altruism appears to exhibit a negative connection with the sensation of possessing a special entitlement to more represents a limitation in terms of drawing broader conclusions about the associations with other psychological facets based solely on the outcomes of this study.

Nevertheless, this study lays the groundwork for potential future explorations. For example, avenues for further inquiry include investigating the interplay between altruism and the altruism scale of Rushton et al. (1981), or developing a scale to gauge the inclination to avoid stark disparities in gains between oneself and others (e.g., "I don't want there to be a disparity in what I receive, whether more or less") and scrutinizing its correlation with equality. Comprehensive investigations into diverse personality traits and their connections with altruism and equality, akin to the works of Haesevoets et al. (2018) and Hilbig et al. (2023), could yield valuable and insightful outcomes.

Whether the estimation of altruism/equality from the SSM data aligns conceptually with that from the EEM data remains a subject of debate. In the context of this study, we arrived at the conclusion that altruism was reasonably consistent across both tasks due to the high correlation observed for altruism. However, the correlation for equality was lower, hinting at the possibility that these two tasks may be capturing distinct qualitative aspects. This divergence could be attributed to the fact that assessing equality within EEM tasks is comparatively more straightforward. The EEM task, in particular, compels participants to actively make choices between equality and inequality, consistently underscoring their awareness of fairness in decision-making. The disparity in the interpretation of equality across these tasks is a topic that warrants further discussion in future research.

6.5 Limitations

Study 2 demonstrated the low retest reliability of the equality parameter. However, the retest reliability of IA was also low, which could be interpreted as a problem with the measurement method or attributed to the possibility that equality is more prone to changes over time than altruism. However, this study did not clarify which explanation is more plausible.

The EEM provides a more accurate estimate, but requires more items, than the SSM, which requires only six items to measure SVO (15 items for measuring IA). Nevertheless, the number of items in the EEM is freely adjustable according to the objectives of the researchers: if they wish to significantly increase the reliability of the EEM, then they can increase the number of items in the EEM by including more details of the money that is presented. On the other hand, if the researchers wish to reduce the burden on the respondents, they may increase the interval between the amounts of money and reduce the number of items. The burden on the respondents could also be reduced by using a format that only requires equal equivalence points. There is a need to further improve the EEM in the future by developing easy-to-use tasks.

6.6 Future perspectives

We believe that the method proposed in this study could lead to cross-disciplinary research on decision-making between psychology and economics. SVO has been conceptualized and primarily used in psychology. On the other hand, there is a similar concept in economics, which is referred to as social preference and encompasses several varieties: type-dependent social preference (Levine 1998), intention-based social preference (Rabin 1993; Falk and Fischbacher 2006), and outcome-based social preference (Fehr and Schmidt 1999; Bolton and Ockenfels 2000). The weights attached to outcomes for self and others in outcome-based social preferences are known as SVO in psychology. SVO and social preferences are very similar, although not identical. How SVO is represented in economic models has not been examined in detail. It has already been demonstrated that the widely recognized economic model for inequality aversion (FS model) and the modified SVO model in this study are mathematically equivalent (Muto 2006). This study assumes parameters corresponding to the FS model. In the FS model, the difference between altruism and equality is the envy parameter, and the sum of altruism and equality is the guilt parameter. Therefore, the altruism and equality parameters can be transformed into the FS model's parameters. This study applied the ideas of reliability and validity in psychometrics to estimate social preference parameters. Evaluating the measurement of social preferences in terms of the reliability and validity of psychology measurements would support future research on social preferences in economics. Therefore, the method proposed in the present study is expected to contribute to the dialogue between SVO researchers in psychology and social preference researchers in economics.

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Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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